Chapter 4. Integers and mathematical induction

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https://github.com/xinhi6716/computer-science-self-study

4.1 Divisibility and well-ordering

For integers a, b, write $a \mid b$ if $b = ak$ for some integer k. Negative divisors are allowed; this relation is not a numerical quotient. In particular,

$$a \mid b,\ a \mid c \Rightarrow a \mid (mb + nc).$$

The well-ordering principle says every nonempty subset of positive integers has a least element. It does not apply directly to every nonempty integer set; all negative integers have no least member. Also $0 \mid b$ holds exactly when $b = 0$. Division arguments usually require a nonzero divisor.

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4.2 The precise form of weak induction

To prove $\forall n \geq n_0, P(n)$, prove $P(n_0)$, then for arbitrary fixed $k \geq n_0$, assume $P(k)$ and derive $P(k+1)$. This covers the successor chain starting at n_0. The base is essential; a skipped or incorrect starting point may invalidate the claim.

For the false claim "all integers $n \geq 2$ are even," moving from $2k$ to $2(k+1)$ covers only even integers. It does not prove the required implication for an arbitrary index. The step may use $P(k)$ repeatedly or split into cases according to k, while keeping k arbitrary and fixed.

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4.2.1 Induction follows from well-ordering

Induction is not confidence gained by checking many successors. Suppose $P(n_0)$ and $P(k) \to P(k+1)$ for every $k \geq n_0$. Let

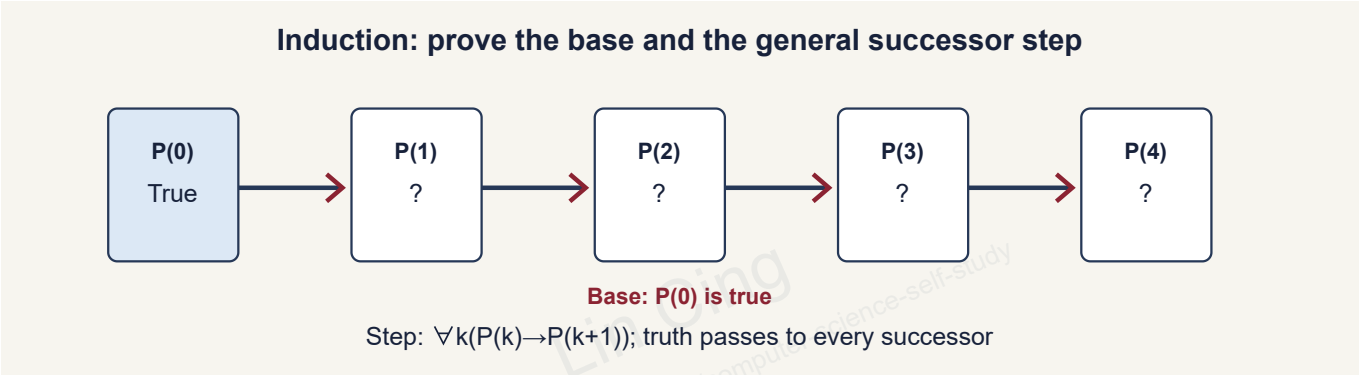
$$S = \{n \in \mathbb{Z} : n \geq n_0 \wedge \neg P(n)\}.$$

If the theorem fails, S is nonempty. Its translate $\{n - n_0 + 1 : n \in S\}$ is a nonempty positive-integer set, so well-ordering gives a least counterexample m. The true base excludes $m = n_0$. Therefore $m - 1 \geq n_0$, and minimality gives $P(m-1)$. The induction step gives $P(m)$, contradicting $m \in S$. Thus S is empty.

Without a base, this argument cannot exclude failure at n_0. Without the step, it cannot pass to the successor.

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4.2.1 Induction follows from well-ordering (continued)



Assuming $P(k)$ is a conditional assumption for one arbitrary fixed index, not an assumption of the whole theorem. The completed step proves an implication.

4.2.2 Complete induction examples and boundaries

For $n \geq 1$, prove

$$1 + 3 + \cdots + (2n - 1) = n^2.$$

At $n = 1$, both sides equal one. Assume the formula at fixed $k \geq 1$. The next term is $2(k + 1) - 1 = 2k + 1$, and

$$1 + 3 + \cdots + (2k - 1) + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2.$$

This is the next-index claim, completing induction.

For $P(n) : 4n < n^2 - 7, n \geq 6$, the base gives $24 < 29$. Assume $4k < k^2 - 7$ with $k \geq 6$. Then

$$4(k + 1) = 4k + 4 < k^2 - 3 < k^2 + 2k - 6 = (k + 1)^2 - 7,$$

where the second inequality uses $2k > 3$. The claim fails at one, since $4 < -6$ is false, so its start cannot be changed to one.

Writing $P(k + 1)$ on scratch paper helps find a route from $P(k)$. The formal proof must start with established assumptions and end at the target, without assuming the target.

4.3 Strong induction and least counterexamples

To prove $P(n)$, strong induction assumes every earlier $P(j)$, $n_0 \le j < n$. It is useful for integer factorization, graph-removal operations, and divide-and-conquer structures. Weak induction is a special case.

A least-counterexample proof forms $S = \{n \ge n_0 : \neg P(n)\}$, assumes it nonempty, and applies well-ordering to derive a smaller counterexample or contradiction. This method differs from contraposition and must explicitly justify nonemptiness.

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4.3.1 Equivalence of weak and strong induction

Assume weak induction valid. Suppose $P(n_0)$ holds and all preceding statements imply $P(n)$ for each $n > n_0$. Set

$$Q(n) = P(n_0) \wedge P(n_0 + 1) \wedge \cdots \wedge P(n).$$

The base $Q(n_0)$ holds. Assuming $Q(k)$ supplies every earlier P; the given strong step yields $P(k+1)$, hence $Q(k+1)$. Weak induction proves every $Q(n)$, and $Q(n) \Rightarrow P(n)$ proves the desired result. Conversely, strong induction includes the most recent hypothesis $P(k)$, so it permits every weak-induction proof. It broadens the usable history without increasing logical proof strength.

For a complete example, let $a_0 = 1, a_1 = 2$, and

$$a_n = 2a_{n-1} + 3a_{n-2} \quad (n \ge 2).$$

Claim $a_n = (3^{n+1} + (-1)^n)/4$. At zero and one, the formula gives $(3+1)/4 = 1$ and $(9-1)/4 = 2$. Fix $k \ge 1$ and assume the formula through k. Then

4.3.1 Equivalence of weak and strong induction (continued)

$$\begin{aligned} a_{k+1} &= 2a_k + 3a_{k-1} \\ &= \frac{6 \cdot 3^k + 3 \cdot 3^k + 2(-1)^k - 3(-1)^k}{4} \\ &= \frac{9 \cdot 3^k - (-1)^k}{4} = \frac{3^{k+2} + (-1)^{k+1}}{4}. \end{aligned}$$

The two-term recurrence needs two bases to start the dependence on $k-1$. Every strong proof can be rewritten as weak induction on a prefix conjunction.

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4.4 Recursive definitions

A recursive definition specifies bases and a rule, such as

$$F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2} \quad (n \geq 2).$$

Existence and uniqueness both need justification. If every new value depends only on smaller indices, finite iteration determines it uniquely. Negative-index references or insufficient initial values can leave a sequence undefined.

More generally, given bases $a_0, \ldots, a_{r-1}$ and a specified deterministic function f,

$$a_n = f(n, a_0, \ldots, a_{n-1}) \quad (n \geq r)$$

defines exactly one sequence. Strong induction constructs it: once all preceding values are uniquely determined, this formula uniquely determines the next one, without a dependency cycle.

For Fibonacci bases $b_0 = 0, b_1 = 1$, iteration gives $b_2 = 1, b_3 = 2, b_4 = 3, b_5 = 5, b_6 = 8, b_7 = 13$. Without the two bases, both $0, 1, 1, 2, \ldots$ and $2, 5, 7, 12, \ldots$ satisfy the same recurrence. Nor does $d_n = d_{n+1}$ without other boundaries uniquely determine an infinite sequence. Recursion describes a definition method; uniqueness depends on well-founded structure and initial conditions.

4.5 Recursive algorithms and structural induction

Recursive factorial needs an explicit base return and calls with strictly smaller arguments remaining inside the domain. Correctness can be proved by induction on argument size.

For $\mathrm{bitsum}(0) = 0$, define for $n > 0$,

$$\mathrm{bitsum}(n) = \mathrm{bitsum}(\lfloor n/2 \rfloor) + (n \bmod 2).$$

The nonnegative argument strictly decreases, so recursion terminates. A trace is

$$\mathrm{bitsum}(45) = 1 + \mathrm{bitsum}(22) = 1 + 0 + \mathrm{bitsum}(11) = \cdots = 4.$$

The full remainder sequence is $1, 0, 1, 1, 0, 1$, matching four ones in $45 = (101101)_2$. For correctness, strong induction assumes all smaller arguments give their binary digit sums, then adds the last binary digit in the displayed recurrence. Termination and correctness are separate claims. Recursion trees or the master theorem analyze complexity.

4.6 The division algorithm

For integer a and positive integer d, unique integers q, r satisfy

$$a = qd + r, \qquad 0 \le r < d.$$

For existence, choose $q = \lfloor a/d \rfloor, r = a - qd$. For uniqueness, two representations give $r - r' = (q' - q)d$, while $-d < r - r' < d$, forcing zero.

Programming-language remainder rules for negative inputs can differ. In particular C/C++ `%` follows its quotient convention rather than the nonnegative mathematical convention for $a \bmod d$.

4.6.1 A well-ordering proof of division

The floor construction is short; well-ordering explains existence of a least nonnegative remainder directly. For fixed $a \in \mathbb{Z}, d > 0$, form

$$S = \{a - dq : q \in \mathbb{Z}, a - dq \geq 0\}.$$

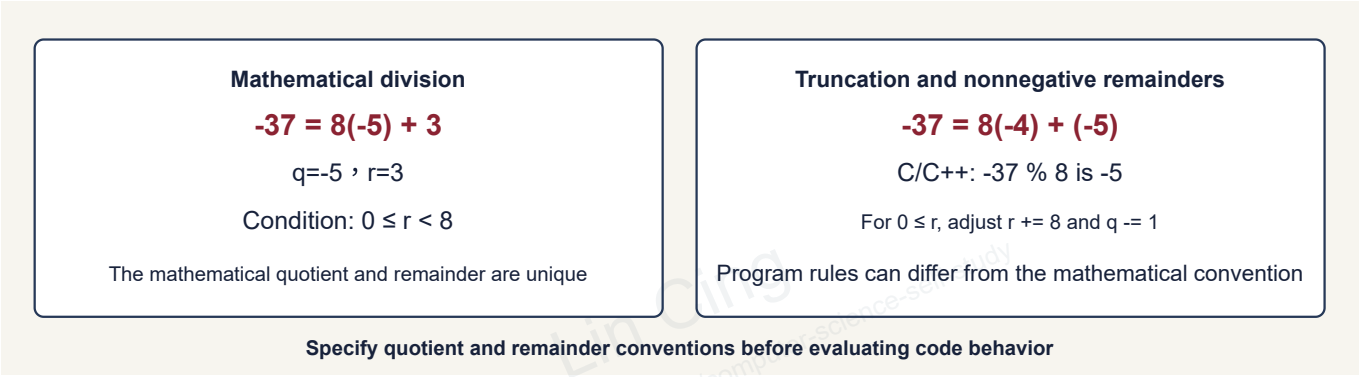
It is nonempty: $q_0 = -|a| - 1$ gives $a + d(|a| + 1) > 0$, since $d \geq 1$. Let its least member be $r = a - dq$. If $r \geq d$, then $r - d = a - d(q + 1)$ is a smaller nonnegative member, contradiction. Thus $0 \leq r < d$.

For another representation $a = dq' + r'$ in the same range,

$$r - r' = d(q' - q), \qquad -d < r - r' < d.$$

Only zero in that interval is divisible by d, so $r = r'$. Since $d > 0$, also $q = q'$.

4.6.1 A well-ordering proof of division (continued)



Exhibiting one pair proves existence but not uniqueness. Excluding a second pair proves uniqueness but not existence.

4.6.2 Base representations and two's complement

For nonnegative integer n, repeated division in integer base $b \geq 2$ gives

$$n = r_0 + r_1 b + \cdots + r_k b^k, \qquad 0 \leq r_i < b.$$

For example,

$$47 = 3 \cdot 15 + 2, \quad 15 = 3 \cdot 5 + 0, \quad 5 = 3 \cdot 1 + 2, \quad 1 = 3 \cdot 0 + 1.$$

Reading remainders upward gives $47 = (1202)_3$, checked by $27 + 18 + 0 + 2 = 47$.

For positive input, each quotient $\lfloor n/b \rfloor < n$, so division terminates at zero. Substitution back proves existence with a nonzero leading digit. Any two expansions have equal last digits modulo b. Subtract that digit, divide by b, and repeat to prove uniqueness. Write zero as one digit 0. Negative integers use a separate minus sign rather than a sum of nonnegative digits times positive powers.

For eight-bit two's complement, write $6 = (00000110)_2$, invert bits to 11111001, and add one:

$$-6 = (11111010)_2.$$

Precisely, w bits represent

4.6.2 Base representations and two's complement (continued)

$$-b_{w-1} 2^{w-1} + \sum_{i=0}^{w-2} b_i 2^i,$$

with range $[-2^{w-1}, 2^{w-1} - 1]$. When the top bit is one, subtract 2^w from unsigned value U. For minus six in eight bits, $U = 256 - 6 = 250$. The top bit has negative weight, rather than being an attached sign label.

Inverting and adding one again recovers 00000110. Fixed-width signed addition may discard the left carry, yet equal-sign operands producing the opposite sign indicate overflow. Eight-bit $127 + 1$ produces 10000000, interpreted as minus 128. Bit patterns and mathematical integer values must be distinguished. Base representation proceeds low digit first, and two's complement requires a fixed width and range convention.

4.7 Primes and irreducibles require full definitions

An integer $p > 1$ is prime when its only positive divisors are one and itself. One is a unit, not prime; zero is not prime either.

In an integral domain R, a prime element p must be nonzero and a nonunit, and satisfy

$$p \mid ab \Rightarrow p \mid a \text{ or } p \mid b \qquad (a, b \in R).$$

A unit has a multiplicative inverse. An irreducible is also nonzero and a nonunit, and every factorization $p = ab$ has a unit factor.

Element	Can the divisibility implication alone hold?	Prime element?
0	Yes in a domain, since $ab = 0$ forces a zero factor	No, nonzero condition fails
Units such as ± 1 in integers	Yes, units divide everything	No, nonunit condition fails
2 in integers	Yes, by Euclid's lemma	Yes, and irreducible
4 in integers	No, it divides $2 \cdot 2$ but neither factor	No; reducible as $2 \cdot 2$

4.7 Primes and irreducibles require full definitions (continued)

A prime is irreducible: if $p = ab$ and $p \mid a$, put $a = pc$ and cancel nonzero p to get $1 = cb$, making b a unit; the other case is symmetric. The converse holds in integers but may fail in other domains. For trial division of positive n, a factorization $n = ab, 1 < a \leq b$ gives $a \leq \sqrt{n}$.

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4.7.1 Prime divisors and infinitely many primes

For composite $n > 1$, its nontrivial divisors $1 < d < n$ form a nonempty set. Let p be the least. If $p = ab$ with $1 < a < p$, then a is a smaller nontrivial divisor of n, contradiction. Thus p is prime. Writing $n = pk$, minimality gives $p \le k$, so $p^2 \le n$. Every composite has a prime divisor at most $\sqrt{n}$, explaining the trial-division stopping condition.

If all primes were $p_1, \dots, p_t$, let

$$N = p_1 \cdots p_t + 1.$$

Since $N > 1$, it has a prime divisor. Every listed prime leaves remainder one on dividing N, so none divides it, contradiction. Thus infinitely many primes exist.

Allowing one as prime would destroy uniqueness through arbitrary factors of one. Zero is also excluded. For $n = 2$, no trial candidates $2 \le d \le \sqrt{n}$ exist, correctly declaring it prime. The test assumes $n > 1$ and does not apply to one.

4.8 GCD and the Euclidean algorithm

For integers a, b not both zero, their gcd is the greatest positive common divisor. For $b > 0$,

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

If $a = qb + r$, the common divisors of a, b are exactly those of b, r. Successive division reduces remainders to zero, leaving the previous positive remainder as gcd.

A zero input returns the absolute value of the other in constant steps. For two nonzero inputs normalized positive, remainders at least halve every two steps. This gives $O(\log \min(|a|, |b|) + 1)$ divisions, not the bit complexity of large-integer arithmetic.

4.8.1 The least positive linear combination

For a, b not both zero, form

$$S = \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\}.$$

If $a \neq 0$, one of $a, -a$ is positive; if $a = 0$, one of $b, -b$ is positive. Thus S is nonempty. Let its least member be $d = ax_0 + by_0$.

Divide $a = dq + r$, $0 \leq r < d$. Then

$$r = a - dq = a - (ax_0 + by_0)q = a(1 - x_0 q) + b(-y_0 q).$$

If $r > 0$, this is a smaller positive member of S, contradiction. Hence $d \mid a$, and similarly $d \mid b$. Every common divisor c divides $ax_0 + by_0 = d$; every positive common divisor is therefore at most d. This proves $d = \gcd(a, b)$ and existence of a Bézout representation.

For Euclidean correctness,

$$d \mid a \text{ and } d \mid b \iff d \mid b \text{ and } d \mid r$$

4.8.1 The least positive linear combination (continued)

follows in the two directions from $r = a - qb$ and $a = qb + r$. The common-divisor set is invariant. The final nonzero remainder is its gcd, and strict remainder descent proves termination.

gcd(1071,462): the preceding remainder becomes the next divisor

Step	Division	Remainder	Invariant
1	1071 = 2 × 462 + 147	147	gcd unchanged
2	462 = 3 × 147 + 21	21	gcd unchanged
3	147 = 7 × 21 + 0	0	Last nonzero remainder = 21

21 = 462 - 3×147 = 7×462 - 3×1071

Thus gcd=21, and 21=s·1071+t·462 has (s,t)=(−3,7).

Calling d the least positive combination does not alone prove maximality. One must show both that it is a common divisor and that all common divisors divide it.

4.9 Bézout and linear Diophantine equations

For a, b not both zero, there are integers s, t with

$$\gcd(a, b) = sa + tb.$$

Extended Euclid retains these coefficients; reversing its division steps constructs them. They may be negative and are not unique.

For $84 = 2 \cdot 30 + 24$, $30 = 24 + 6$,

$$6 = 30 - 24 = 30 - (84 - 2 \cdot 30) = 3 \cdot 30 - 84.$$

Thus $84x + 30y = 12$ is solvable, since six divides twelve. Multiply by two to obtain $-2 \cdot 84 + 6 \cdot 30 = 12$, with $(x, y) = (-2, 6)$. If $d = sa + tb$ and $g = \gcd(a, b)$, then for every integer k,

$$d = (s + k\,b/g)a + (t - k\,a/g)b.$$

More generally, for a, b not both zero,

$$ax + by = c$$

4.9 Bézout and linear Diophantine equations (continued)

has integer solutions iff $g = \gcd(a, b)$ divides c. If $c = g\ell$, multiply $g = sa + tb$ by ℓ to construct a solution. Conversely, every common divisor divides the left side of any solution, hence divides c.

Handle $a = b = 0$ separately: every integer pair solves it when $c = 0$, and none when $c \neq 0$. This convention leaves $\gcd(0, 0)$ undefined. Chapter 14 uses Bézout to test modular invertibility.

4.9.1 GCD, LCM, and common multiples

For positive integers a, b, the lcm is their least positive common multiple. We prove

$$\gcd(a,b)\,\operatorname{lcm}(a,b) = ab.$$

Write $a = gA, b = gB$, where $g = \gcd(a,b)$ and $\gcd(A,B) = 1$. The candidate $L = gAB = aB = bA$ is a common multiple.

Any positive common multiple has $M = au = bv$, giving $Au = Bv$. Bézout gives $sA + tB = 1$. Multiply by v: both terms on the left are divisible by A, the second because $Bv = Au$. Thus $A \mid v$; write $v = Aw$. Then $M = gB(Aw) = Lw$, so L divides every positive common multiple and is least. Finally $gL = g^2AB = ab$.

For $\gcd(168, 456) = 24$,

$$\operatorname{lcm}(168, 456) = 168 \cdot 456/24 = 3192.$$

Indeed $168 = 24 \cdot 7, 456 = 24 \cdot 19$, with coprime 7 and 19. Every common multiple must include their product after removing the common factor.

4.9.1 GCD, LCM, and common multiples (continued)

The displayed identity uses positive inputs. Other integer, sign, and zero conventions must be specified first. For example $\gcd(0, n) = |n|$, while $\operatorname{lcm}(0, n)$ requires its own convention.

4.9.2 Prime and irreducible can differ

In a commutative integral domain with identity, both definitions require a nonzero nonunit. Prime means $p \mid ab \Rightarrow p \mid a$ or $p \mid b$; irreducible means $p = ab \Rightarrow$ at least one factor is a unit.

For prime $p = ab$, primality gives $p \mid a$ or $p \mid b$. In the first case $a = pc$ gives $p = pbc$; cancel nonzero p in the domain to obtain $1 = bc$, so b is a unit. The other case is symmetric.

In integers the converse holds. For irreducible $p > 1$ with $p \mid ab, p \nmid a$, its only positive divisors imply $\gcd(p, a) = 1$. Bézout gives $up + va = 1$. Multiply by b: $upb + vab = b$. Both terms are divisible by p, giving $p \mid b$.

The converse fails in

$$R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}.$$

Define $N(a + b\sqrt{-5}) = a^2 + 5b^2$. Multiplication expands to give $N(uv) = N(u)N(v)$; nonzero elements have positive integer norms. Norm one occurs exactly at ± 1, precisely the units, since inverse norms must multiply to one.

4.9.2 Prime and irreducible can differ (continued)

If $3 = uv$ with both factors nonunits, their norms multiply to nine and must both equal three. But $a^2 + 5b^2 = 3$ has no integer solution: nonzero b gives at least five, and $b = 0$ requires $a^2 = 3$. Thus three is irreducible. It is false that all nonzero norms below nine are units; two has norm four and $\sqrt{-5}$ norm five.

Yet

$$(2 + \sqrt{-5})(2 - \sqrt{-5}) = 9$$

is divisible by three, while neither factor is: comparing integer coefficients would require three to divide two and ± 1. Three is not prime in this domain. The integer equivalence cannot be transferred to arbitrary domains. Omitting nonzero and nonunit conditions wrongly admits zero or units into the definitions.

4.10 The fundamental theorem of arithmetic

Every integer $n \geq 2$ is a product of primes, uniquely up to order. For existence, the base two is already prime. Assume all integers from two to $n-1$ factor. If n is prime, stop. Otherwise $n = ab$ with $1 < a, b < n$. Strong induction factors both, and concatenation factors n.

For uniqueness, first prove Euclid's lemma without using unique factorization. If prime $p \mid ab$ and $p \nmid a$, then $\gcd(p, a) = 1$, so Bézout gives $up + va = 1$. Multiplication by b gives $upb + vab = b$. Both left terms are divisible by p, so $p \mid b$.

Extend the lemma to any finite product by induction on its length. The one-factor case is immediate. If $p \mid (a_1 \cdots a_m)a_{m+1}$, the lemma gives divisibility of the last factor or the first product. In the latter case, the induction hypothesis supplies a divisible factor among the first m.

Now suppose

$$n = p_1 \cdots p_r = q_1 \cdots q_s$$

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4.10 The fundamental theorem of arithmetic (continued)

are prime factorizations. The extended lemma makes p_1 divide some q_j, so the two positive primes are equal. Reorder and cancel them, then repeat. If one list ends before the other, a remaining product of primes greater than one would equal one, impossible. Thus the prime lists match with multiplicities, proving uniqueness up to order. In a general integer-ring formulation, specify association up to multiplication by a unit.

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4.10.1 Standard form, divisor counts, and squares

Write the unique factor multiplicities as

$$n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}, \qquad \alpha_i \geq 1.$$

Every positive divisor uniquely has $d = p_1^{\beta_1} \cdots p_r^{\beta_r}, 0 \leq \beta_i \leq \alpha_i$. Independent exponent choices and uniqueness give

$$\tau(n) = \prod_i (\alpha_i + 1).$$

A divisor is a square iff all its exponents are even, so square divisors number

$$\prod_i \left(\left\lfloor \frac{\alpha_i}{2} \right\rfloor + 1 \right).$$

For $360 = 2^3 \cdot 3^2 \cdot 5$, $\tau(360) = 4 \cdot 3 \cdot 2 = 24$. Square exponents for two are 0,2; for three 0,2; and for five only zero. The four square divisors are $1, 4, 9, 36$.

If $n = m^2$, every prime exponent is even; conversely halving even exponents constructs a square root. Also $\tau(n)$ is odd iff every $\alpha_i + 1$ is odd, equivalently every α_i even, so iff n is a square.

4.11 Proof defects and boundaries

"Every $n \geq 2$ is a product of two integers greater than one" is false at every prime. The corrected conditional for nonprime n needs a precise composite definition.

A strong-induction factorization proof cannot invoke $P(1)$ before establishing its actual base $n = 2$. Every recursive reference must lie in an established or strictly smaller part of the domain.

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4.12 Checks and exercises

Check the quantified starting point, bases, decreasing measure, uniqueness of division, gcd invariant, and assumptions for prime-factorization uniqueness. The separate exercise PDF provides questions, and the solution PDF gives their derivations.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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